

## Supporting Information

### Null-Model Choice for Rule-Defined Pattern Counts in Clustered Longitudinal Sequences: A Case Study of Jewish Halachic Menstrual Anticipation Rules

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This document contains the exploratory between-woman pairwise dependence analysis referred to in Sections 3.4, 4.4 and 5.2 of the main article (Section S1), the literature search referred to in Section 1 (Section S2) and the evaluation of the zero-count rule *Haflaga Chozar Chalila* referred to in Sections 2.1 and 3.1 (Section S3). The pairwise analysis is secondary to the null-model comparison and does not bear on the primary question of whether within-woman ordering produces excess rule-defined events. Notation follows the main article; all computations are produced by `scripts/postprocess.py` and the accompanying notebook in the code repository (<https://github.com/dvirross/VesetVectorTest>).

The SHA-256 checksum of the released filtered data file (`data/FilteredData.csv`; 1,554 cycles, 118 women), for verifying file integrity, is

508ee88efb18bcd29c7ed6f841827377bb8d72a8c7dd38fdeb40cf08691f7811

## S1. Exploratory Pairwise Dependence Between Pattern Types

### S1.1 Question and expectations

The joint test of the main article treats the five counts as one vector. The pairwise analysis asks a different and secondary question — whether women who produce one pattern type tend to produce another — as a consistency check against the relationships derived from the pattern definitions (main article, Section 2.2). Those relationships lead to two expectations: a positive  $\text{Haflaga} \times \text{Week-Dilug}$  association, which is partly an arithmetic consequence of containment (every *Haflaga* sequence at  $H = 30$  contains a *Week-Dilug* event) rather than an empirical hypothesis, and a plausible but not entailed positive  $\text{Haflaga} \times \text{Dilug-in-Dilug}$  association from polynomial proximity. No expectation is held for the other eight pairs, so the analysis is a consistency check, not a search.

## S1.2 Methods and rationale

**Unit of analysis: the woman.** — Pattern events are defined within a woman’s sequence and cycles within a woman are not independent, so the woman is the only exchangeable unit for an association test. The analysis works with 118 per-woman summaries, not with 1,554 cycles.

**Rates rather than counts.** — A woman with 45 cycles has roughly nine times as many opportunities to produce *any* pattern as a woman with 5 cycles, so raw per-woman counts of two patterns are positively correlated through follow-up length alone. Dividing each count by the number of cycles removes the first-order part of this exposure effect. It does not remove it entirely: a rate from 5 cycles is far noisier than one from 45, the minimum run lengths (2, 3 or 4 cycles) mean short sequences have proportionally fewer eligible windows, and in these data the Haflaga rate itself correlates with follow-up (Spearman correlation with  $n_i$ : 0.25 for Haflaga, 0.13 for Week, 0.10 for Dilug-in-Dilug, 0.02 for Dilug, 0.00 for Week-Dilug). The permutation null below is therefore one of *marginal* independence between two rates, not of independence conditional on follow-up; Section S1.4 checks how much this matters.

**Spearman rather than Pearson correlation.** — Per-woman rates are bounded at zero, heavily zero-inflated (most women have no event of a given type) and right-skewed. Pearson’s coefficient would be dominated by the few women with several events; the rank correlation measures monotone association and is insensitive to the magnitude of those few values.

**Permutation rather than asymptotic  $p$ -values.** — With 118 women, many tied zeros and skewed rates, the usual  $t$ -approximation for the null distribution of  $r_s$  is unreliable. One rate vector is permuted across women 50,000 times (seed 17); this breaks any association between the two patterns while preserving both marginal distributions exactly, giving a two-sided Monte Carlo  $p$ -value (proportion of permutations with  $|r_s^*| \geq |r_s|$ , with the  $(b+1)/(B+1)$  convention).

**Holm correction over ten pairs.** — Five patterns yield ten pairs, of which only two carry a structural expectation. Holm’s step-down procedure controls the family-wise error rate at  $\alpha = .05$  over all ten and is valid under any dependence among the tests, which matters because the pairs share patterns.

**Binary robustness check.** — Because the rate metric depends on the exposure normalisation and on the few women with several events, the analysis is repeated with each woman coded 1/0 for ever having established each pattern. Each pair is then a  $2 \times 2$  table tested with Fisher’s exact test (conditional on the margins) and with a permutation  $\chi^2$  statistic (permuting one indicator across women), again Holm-corrected. The probability of ever observing an event also increases with follow-up, so this check addresses the choice of metric, not the follow-up dependence.

### S1.3 Results

Table S1 and Figure S1 give the rate-based Spearman correlations. Haflaga  $\times$  Dilug-in-Dilug is positive and survives Holm correction ( $r_s = 0.27$ ,  $p_{\text{perm}} = .004$ , Holm  $p = .038$ ): women with a higher rate of exact repetitions also tend to have a higher rate of second-order progressions, most of which ( $|d| = 1$  in 16 of 27) are near-constant sequences. Haflaga  $\times$  Week-Dilug is positive, as containment requires, but does not survive correction ( $r_s = 0.23$ ,  $p_{\text{perm}} = .011$ , Holm  $p = .10$ ); only 3 of the 26 Haflaga sequences are at  $H = 30$ , so the guaranteed part of the association is small. The remaining eight pairs are null (all Holm  $p = 1$ ), including Dilug  $\times$  Week-Dilug, whose weak negative sign ( $r_s = -0.12$ ) is in the direction implied by their mutual exclusion but is not distinguishable from zero.

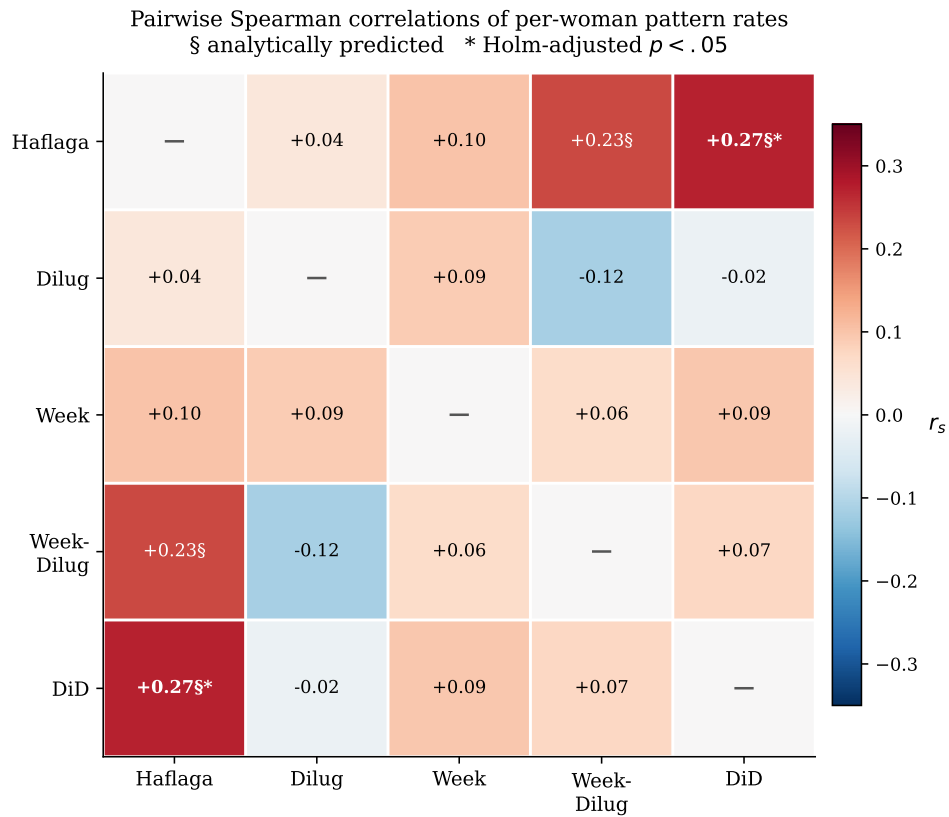
The binary (ever/never) version reproduces the ordering and the two smallest  $p$ -values (Haflaga  $\times$  Dilug-in-Dilug:  $\chi^2 = 9.99$ ,  $p_{\text{perm}} = .004$ , Holm  $p = .042$ , Fisher  $p = .004$ ; Haflaga  $\times$  Week-Dilug:  $\chi^2 = 7.26$ ,  $p_{\text{perm}} = .013$ , Holm  $p = .12$ ), so the findings do not depend on the rate normalisation or on the few women with multiple events.

### S1.4 Robustness to follow-up length

Because both rates in a pair can depend on follow-up, the unrestricted permutation tests marginal rather than follow-up-conditional independence. Two checks, computed by the same script with the same seed, bound the consequence for the two structurally expected pairs. First, the partial Spearman correlation given  $n_i$  (correlation of the rank residuals after regressing each rank vector on the rank of  $n_i$ ) is 0.254 for Haflaga  $\times$  Dilug-in-Dilug (unadjusted 0.269) and 0.239 for Haflaga  $\times$  Week-Dilug (unadjusted 0.231). Second, restricting the 50,000 permutations to

**Table S1:** Pairwise Spearman correlations between per-woman pattern rates ( $N = 118$ ), sorted by  $p_{\text{perm}}$  (two-sided, 50,000 permutations);  $p_{\text{Holm}}$  across the 10 pairs. §: pair with an analytically expected positive association (main article, Section 2.2). Bold:  $p_{\text{Holm}} < .05$ .

| Pair                                               | $r_s$        | $p_{\text{perm}}$ | $p_{\text{Holm}}$ |
|----------------------------------------------------|--------------|-------------------|-------------------|
| <b>Haflaga <math>\times</math> Dilug-in-Dilug§</b> | <b>+0.27</b> | <b>.004</b>       | <b>.038</b>       |
| Haflaga $\times$ Week-Dilug§                       | +0.23        | .011              | .102              |
| Dilug $\times$ Week-Dilug                          | −0.12        | .206              | 1.00              |
| Haflaga $\times$ Week                              | +0.10        | .277              | 1.00              |
| Week $\times$ Dilug-in-Dilug                       | +0.09        | .326              | 1.00              |
| Dilug $\times$ Week                                | +0.09        | .337              | 1.00              |
| Week $\times$ Week-Dilug                           | +0.06        | .476              | 1.00              |
| Week-Dilug $\times$<br>Dilug-in-Dilug              | +0.07        | .480              | 1.00              |
| Haflaga $\times$ Dilug                             | +0.04        | .668              | 1.00              |
| Dilug $\times$ Dilug-in-Dilug                      | −0.02        | .819              | 1.00              |



**Figure S1:** Spearman correlations between per-woman pattern rates. §: analytically expected positive pair; \*: Holm-adjusted  $p < .05$ . All pairwise tests are exploratory.

exchange rates only among women in the same follow-up quintile (strata of 24, 8, 31, 27 and 28 women) preserves each rate's dependence on  $n_i$  while breaking the association between patterns; the resulting two-sided  $p$ -values are .0046 for Haflaga  $\times$  Dilug-in-Dilug (unrestricted .0036) and .0097 for Haflaga  $\times$  Week-Dilug (unrestricted .0113). The exploratory conclusions are therefore not an artefact of unequal follow-up. These checks are reported as robustness diagnostics for an exploratory analysis, not as an additional primary result.

## S1.5 Interpretation

Three cautions apply. The associations are between women, not within sequences, and say nothing about temporal ordering. They involve rare patterns (18 Week-Dilug and 27 Dilug-in-Dilug events) and rest on few women. And the surviving association is compatible with the proximity argument but is not evidence for it: attributing co-occurrence to a mechanism would require an explicit generative model for the sequences, which the article does not fit.

## S2. Literature Search for Prior Randomisation-Based Tests

The main article states that we are not aware of any formal randomisation-based test of whether the frequency of vessel-type arithmetic patterns in empirical cycle data departs from a chance null. A hit was taken to contradict this only if it used empirical cycle data, defined rule-based arithmetic events (equal intervals, constant first or second differences, weekly anchors or equivalent) and compared their observed frequency with a randomisation, permutation, bootstrap or Monte Carlo null. Descriptive halachic studies, forecasting models, Markov models and regularity statistics were not counted.

The search had two stages. First, an unstructured search of the Hebrew and English halachic, medical-halachic and biostatistical literatures carried out by the author while preparing the dissertation (2019–2022); the quantitative halachic works it located are cited in Section 2.3 of the main article and none tests pattern frequencies against a randomisation null. Second, a logged web search on 16 September 2026 of twelve queries in English and Hebrew combining the halachic terms (*vessel*, *veset*, *haflaga*, *dilug*, *onah*, *niddah*) with statistical terms (permutation, randomisation, Monte Carlo, probability, empirical data), together with citation-chasing queries

for the author’s earlier articles and for re-analyses of the Marquette dataset. Every returned item was screened by title and snippet. No publication meeting the criterion was found in either stage. The verbatim queries, hit descriptions and limitations of the search (general web index rather than bibliographic databases; snippet-level screening; possible unindexed Hebrew material) are recorded in `revision_reports/literature_search_log.md` in the code repository. The main article therefore retains the wording “we are not aware of” rather than asserting absence.

### S3. The Zero-Count Rule *Haflaga Chozer Chalila* Under the Three Nulls

**Definition.** — *Haflaga Chozer Chalila* (“recurring haflaga”) is established when a block of  $m \geq 3$  consecutive haflaga values is repeated exactly, three times in succession, so that the event occupies a window of  $3m \geq 9$  cycles. Thus (25, 27, 26, 25, 27, 26, 25, 27, 26) is an event, (25, 27, 26, 28) repeated three times is an event with  $m = 4$ , and (25, 27, 25, 27, 25, 27) is not (block of length 2). It is the sixth and last rule computable from interval lengths alone; the other repeating rule of the halachic literature, *Dilug Chozer Chalila*, depends on calendar dates and is not computable from these data.

**Counting convention.** — As for the five primary rules, distinct maximal  $m$ -periodic segments of length at least  $3m$  count once each. A segment whose block has a shorter period  $d$  dividing  $m$  is not counted at  $m$ : it is the same event at period  $d$  if  $d \geq 3$ , a Haflaga if  $d = 1$ , and not an event if  $d = 2$ . The rule is implemented twice (a reference loop and a vectorised version) in `scripts/hcc_rule.py`, and the two implementations agree on the observed data and on 3,000 random sequences with planted repeating blocks. In the observed data 100 of the 118 women have the nine or more cycles needed for an event; the observed count is 0.

**Null distributions.** — The rule was counted under  $H_G$ ,  $H_{iid}$  and  $H_W$  with  $B = 50,000$  replicates each (seed 17), using the same generators as the main analysis (Table S3). Under the two global nulls an event occurs in fewer than 1 in 2,000 replicates; under  $H_W$ , which preserves each woman’s multiset of values, a periodic reordering of a woman with few distinct values and many cycles is occasionally produced, and at least one event occurs in 2.8% of replicates. The

observed zero is the modal outcome under every null and its two-sided Monte Carlo  $p$ -value is 1. The replicate counts are stored in `results/hcc_null.npz`.

**Effect on the joint statistic.** — Because a near-degenerate component could in principle disturb the estimated covariance of the joint test, the check was made directly. The three null streams were regenerated with the seed and call order of the main analysis, so that the five primary counts reproduce the stored replicates exactly, and the rule was counted on the same permuted sequences, giving paired six-component replicate vectors (`scripts/hcc_joint.py`; `results/hcc_joint.npz`). Table S2 compares the five- and six-component Mahalanobis tests. The observed  $D_M$  is unchanged to two decimals under every null, because the observed deviation of the sixth component from its null mean is essentially zero. The  $p$ -values increase slightly: every replicate that contains an event lies far beyond the observed vector in the sixth coordinate, so such replicates are added to the exceedance count (16, 19 and 1,378 replicates under  $H_G$ ,  $H_{iid}$  and  $H_W$ ). The split-batch estimate agrees. Under the two global nulls the joint test still rejects at  $p < .001$ ; under  $H_W$  the  $p$ -value moves from .126 to .150 and the test still does not reject. The condition numbers of the estimated  $6 \times 6$  covariance matrices ( $1.4 \times 10^5$ ,  $1.2 \times 10^5$  and  $2.0 \times 10^3$ ) reflect the near-zero variance of the sixth component but caused no numerical difficulty.

**Table S2:** Joint Mahalanobis test with five components (main article) and with *Haflaga Chozer Chalila* added as a sixth component, on the same  $B = 50,000$  paired replicates per null.

| Null      | $D_M^{(5)}$ | $p^{(5)}$ | $D_M^{(6)}$ | $p^{(6)}$ | Split-batch<br>$D_M^{(6)}, p^{(6)}$ |
|-----------|-------------|-----------|-------------|-----------|-------------------------------------|
| $H_G$     | 5.08        | .0001     | 5.08        | .0005     | 5.07, .0005                         |
| $H_{iid}$ | 4.87        | .0004     | 4.87        | .0007     | 4.87, .0007                         |
| $H_W$     | 2.93        | .126      | 2.94        | .150      | 2.96, .143                          |

**Table S3:** *Haflaga Chozer Chalila* under the three nulls ( $B = 50,000$  each; observed count 0).

| Null      | Mean   | SD    | Max | $P(\text{count} \geq 1)$ | $p$<br>(two-sided) |
|-----------|--------|-------|-----|--------------------------|--------------------|
| $H_G$     | 0.0003 | 0.018 | 1   | 0.0003                   | 1.00               |
| $H_{iid}$ | 0.0004 | 0.020 | 1   | 0.0004                   | 1.00               |
| $H_W$     | 0.028  | 0.166 | 2   | 0.028                    | 1.00               |